

RADIO GALAXY

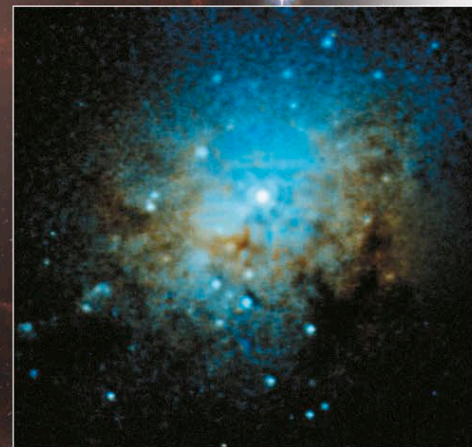
NGC 1275



CATALOG NUMBER	NGC 1275
SHAPE	Elliptical and distorted spiral
DISTANCE	235 million light-years
DIAMETER	70,000 light-years
MAGNITUDE	11.6

PERSEUS

Despite being cataloged as a Seyfert galaxy by Carl Seyfert himself (see p.323), NGC 1275 has remained a mystery. Recent observations have shown that there are two objects—one in front of the other. A ghostly spiral galaxy, revealed by its bright blue star clusters, is responsible for the dust lanes that cross the bright central region, but this brighter region is in fact a separate galaxy. Despite its Seyfert-like core, it is an elliptical, not a spiral. This galactic giant lies at the heart of the Perseus galaxy cluster, and the foreground spiral is racing toward it at 6.7 million mph (10.8 million kph), its structure already disrupted by the elliptical's gravity. Adding to the complexity, the elliptical galaxy is also a radio source, and some astronomers have argued that it shows blazarlike activity (see BL Lacertae, opposite). Whatever the details, NGC 1275 displays many of the typical features of an active galactic nucleus.



CLUSTERS IN THE NUCLEUS

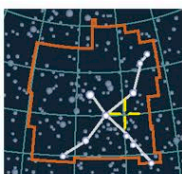
The core of NGC 1275 offers clues to the origin of globular clusters—numerous globularlike clusters are found here, but they are composed of young blue rather than old yellow stars.

ATYPICAL ELLIPTICAL GALAXY

NGC 1275 is unusual for an elliptical galaxy in having a Seyfert-like core. The dark dust lanes are the remains of a now-disrupted separate spiral galaxy in front of NGC 1275.

RADIO GALAXY

Cygnus A



CATALOG NUMBER	3C 405
SHAPE	Pec (peculiar)
DISTANCE	600 million light-years
DIAMETER	120,000 light-years (excluding radio lobes)
MAGNITUDE	15.0

CYGNUS

The most spectacular and powerful radio galaxy in the nearby universe, Cygnus A was discovered as soon as radio telescopes began operating in the 1950s. It features two huge lobes of material emitting radio waves. The lobes are visibly linked to their origin at the heart of a faint, central, elliptical galaxy by two long, narrow jets. From lobe to lobe, the entire structure extends over half a million light-years. Despite its prominence in the radio sky, mysteries still surround Cygnus A, largely because of its great remoteness. Early observations led astronomers to believe the central galaxy was in fact a pair of colliding galaxies. Hubble Space Telescope images suggested a resemblance to NGC 5128, the Centaurus A Galaxy



(see p.322), which is thought to be an elliptical galaxy that has recently swallowed a spiral. Recent detection of a large cloud of red-shifted gas moving through the Cygnus A Galaxy suggests that a collision may indeed be the root cause of the activity. Astronomers have also argued about the origin of the “hot spots,” where the radio lobes glow brightest

at either end. Studies by the Chandra X-ray Telescope have shown that Cygnus A lies at the center of a cloud of hot but sparse gas. The jets have blown out a football-shaped cavity in the gas so vast that it dwarfs the central galaxy. Tendrils of gas, which are emitting X-rays and radio waves, are also falling back down through the cavity onto the poles of the galaxy,

COMPLEX SOURCE

This composite image shows Cygnus A's radio jets in red, X-ray emission in blue, and visible light in yellow. The central galaxy is surrounded by a huge bubble of hot gas through which the radio jets burst out. drawn by its gravitational pull. The hot spots are apparently created where the outward blast of the jets collides with the hot gas falling inward.